ECO 331

Economic History

Technology, Labor, and Inequality

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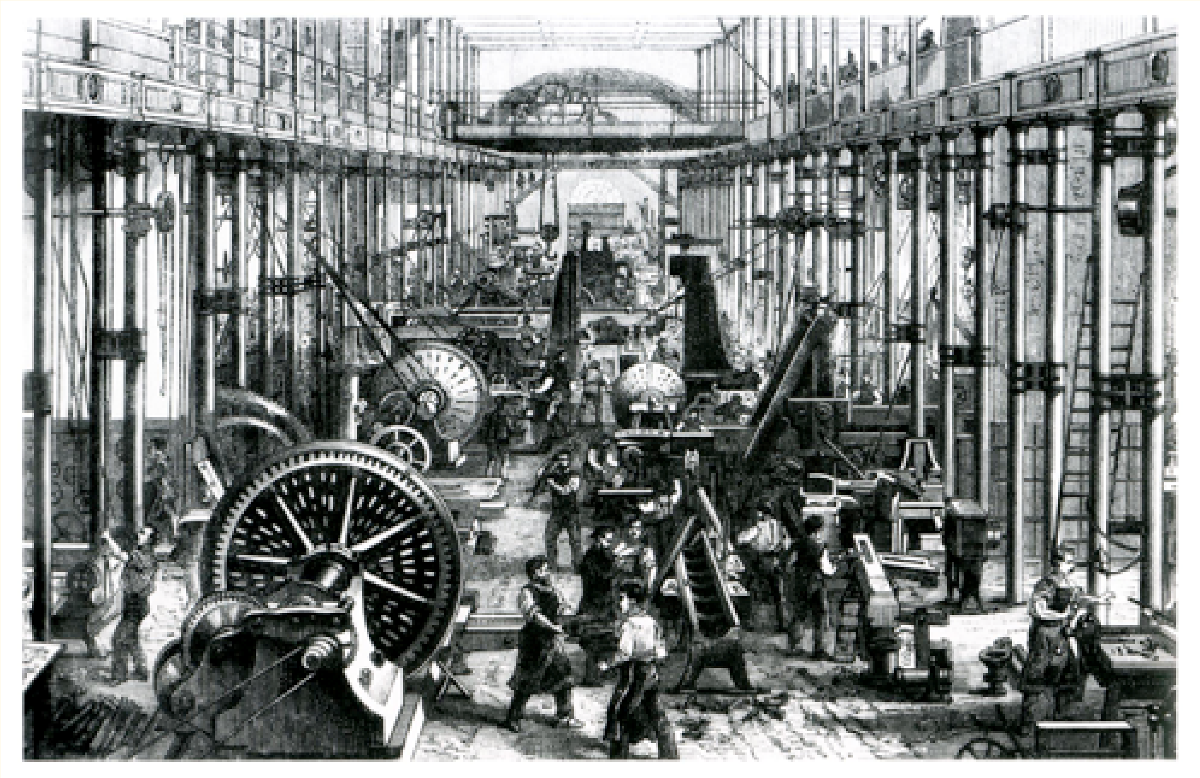
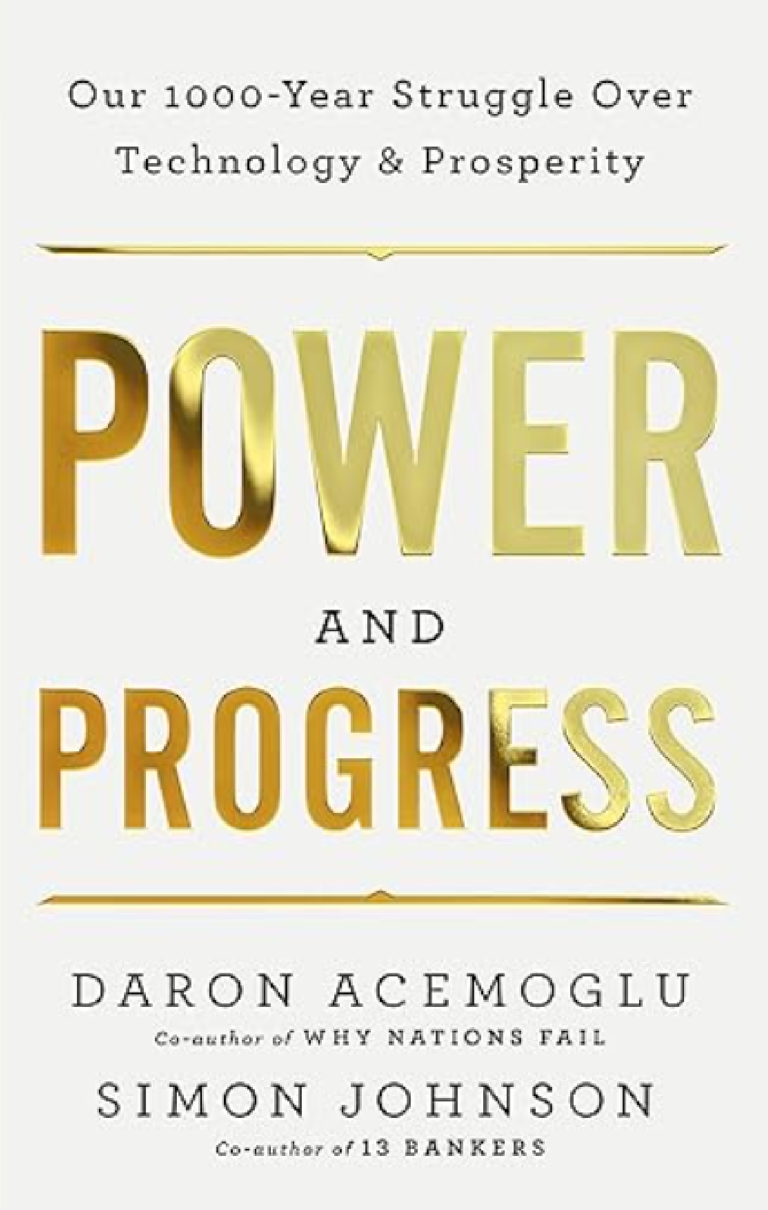


Technology and Labor in Historical Perspective

- Over the semester, we have examined the British Industrial Revolution, American frontier settlement, slavery, and the East Asian economic expansions.
- A central question connects these transitions: Does technological change primarily displace workers, or does it generate new employment?
- The historical record suggests the outcome depends heavily on the institutional framework.
- Today, we use the Acemoglu and Restrepo (2019) framework to connect these historical cases to the modern automation and AI debate.

The Direction of Technological Change

- Tech change is not neutral, inevitable force.
- The *direction* of innovation—whether it saves labor or creates new industries—is shaped by incentives.
- Determinants include:
  - **Factor Endowments:** The relative scarcity and cost of land, labor, and capital.
  - **Market Environment:** Scale and elasticity of demand.
  - **Institutions:** Property rights, tax policy, labor laws, bargaining power of labor and capital.



The Task-Based Model of Production

- Standard macroeconomic models treat production as a function of aggregate capital and labor: $Y = F(K, L)$.
- In that framework, adding capital generally makes labor more productive.
- Acemoglu & Restrepo argue this misses the mechanics of the shop floor. They model production as a **continuum of discrete tasks**.
- **Automation** occurs when capital specifically replaces labor in a given task.

Three Competing Forces

1. **Displacement Effect:** Automation pushes labor out of specific tasks, reducing the labor share of value and placing downward pressure on wages.
2. **Productivity Effect:** Machines lower production costs, leading to lower prices and increased overall output. This raises demand for labor in non-automated tasks.
3. **Reinstatement Effect:** Technological change can also create **new tasks** where humans hold a comparative advantage, pulling labor back into production.

The Race Between Man and Machine

Automation Threshold (I) 50 Tasks

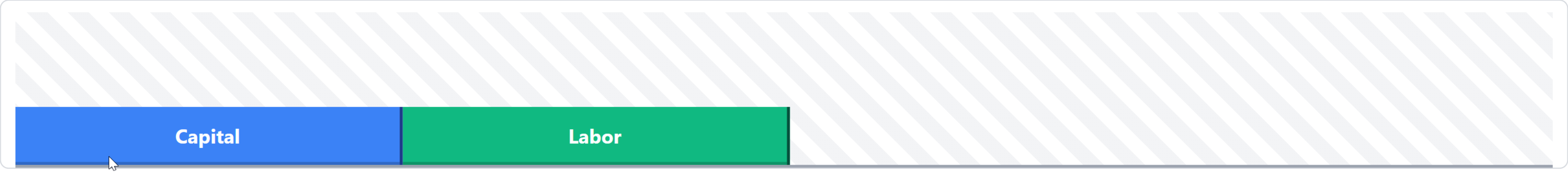
Moves the boundary right, displacing labor.

New Tasks (N) 100 Tasks

Expands the frontier right, reinstating labor.

Capital Productivity (A^K) 1.0x

Makes existing machines more efficient (Intensive margin).



Labor Task Share (Γ)	Total Output (Y)	Wage Bill (Γ × Y)
0.50	100	50

Contrasting tech regimes

Regime	Automation I	New Tasks N	Capital Prod. A^K	Task Share Γ	Total Output Y	Wage Bill $\Gamma \times Y$
Benchmark	50	100	1.0x	0.50	100	**50**
"So-So" Tech	80	100	1.0x	0.20	100	**20**
"Brilliant" Auto	60	100	2.5x	0.40	190	**76**
Golden Age	80	160	1.5x	0.50	200	**100**

Note: Automation (I) moves the threshold right; New Tasks (N) expands the frontier; A^K multiplies machine efficiency.

1. Britain: Displacement and the Handloom Weavers

- How does this model apply to the early British Industrial Revolution?
- **The Displacement Shock:** The introduction of the power loom and spinning frame directly threatened the task share of highly skilled, well-paid artisans.
- The Luddite movement (1811–1816) was not an irrational rejection of progress; it was a targeted response by textile workers experiencing a rapid, severe displacement effect that eroded their human capital and bargaining power.

State Intervention and Labor's Bargaining Power

- Why were the weavers ultimately unable to capture the productivity gains of mechanization?
- The British state actively intervened to support capital owners and suppress labor organization.
- **The Combination Acts (1799/1800)** criminalized trade unionism and collective bargaining.
- **The Frame Breaking Act (1812)** made the destruction of mechanized looms a capital offense.
- **The Result:** With labor legally restrained and a growing supply of displaced rural workers, capital captured the productivity gains. Wages stagnated for decades ("Engels' Pause").

Automation of Manufacturing (US, 1899)

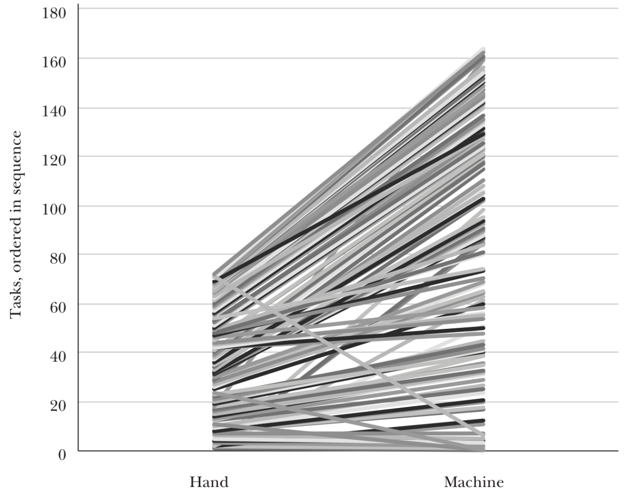
- To observe these task shifts at the micro-level, *Atack, Margo, and Rhode (2019)* use the 1899 "Hand and Machine Labor" study.
- This study documented production methods before and after factory mechanization in the United States.
- It focused strictly on the **task level**: what specifically does a worker do, and how long does it take?
- A primary example: Making 100 pairs of medium-grade shoes.

The Shift to Extreme Specialization

- Under **hand production**: An artisan performed most tasks involved in making a shoe.
- The median number of tasks per worker was 2 (and often much higher for masters).
- Under **machine production**: The median tasks per worker fell to **1**.
- Workers became highly specialized, allowing steam-powered machines to take over routine physical steps.

Figure 1

Slope Chart Linking Hand to Machine Tasks for Unit 71



Source: Authors.

Note: Figure 1 relates each of the various hand tasks for shoe-making on the left to the (far more numerous) machine tasks for shoe-making on the right. Tasks are numbered in sequence. Some hand tasks link to multiple machine tasks. Some are performed in quite different sequences between hand and machine production—these lines cross over. A few hand tasks like "selecting and sorting stock" vanish in machine production (we have connected these to "Task 0" in machine production on the right-hand side). Moreover, the white space on the right-hand axis to which no hand production tasks connect represent new tasks created by mechanization for which there was no hand production analog.

the production of the product (and its supervision). Consequently, any analysis of productivity, including ours, must rely on the measure provided by the study—the amount of time that it took to complete a task—rather than a measure that would be more conventional for economists like value added per worker. Third, while the agents recorded additional information on the survey form that would have been very useful to have for some analyses—for example, the names of the individual workers, and the address of the establishment—this information was not included in the published study. Moreover, as far as we can determine, the completed survey forms have not survived and so this additional information has been lost.[5] Finally,

[5] We tracked down copies of the original survey instrument which are now stored in Record Group (RG) 257 in the US National Archives (US Bureau of Labor Statistics 1890–1905). The forms asked for additional information that was not published, such as the name and location of the establishment, and the names of the workers employed in the production of the various articles.

Task Transitions and Reinstatement

- Mechanization subdivided complex tasks and consolidated others.
- Crucially, it also generated **new tasks** (The Reinstatement Effect).
- Approximately one-third of the tasks in machine production were new to the process.
- Examples included maintaining steam engines, quality inspection, and specialized foreman supervision.

Table 1
Tasks Transitions, Hand to Machine Labor

Transition	*Share of tasks, equal weights*	*Share of tasks, time weights*	*Share using steam power, equal weights*	*Share using water power, equal weights*	*Share using steam power, time weights*	*Share using water power, time weights*
A: Hand Labor						
1:0	0.044	0.030	0.003	0.002	0.002	< 0.001
1:1	0.673	0.604	0.014	0.017	0.009	0.020
1:*M*	0.134	0.192	0.023	0.005	0.008	0.006
N:*M*, *N* > 1, *M* > 1	0.040	0.054	< 0.001	0.010	< 0.001	0.003
N:1	0.108	0.121	0.010	0.018	0.031	0.019
Total	1.000	1.000	0.014	0.014	0.011	0.016
B: Machine Labor						
1:1	0.458	0.563	0.436	0.029	0.461	0.033
1:*M*	0.146	0.172	0.558	0.058	0.538	0.068
N:*M*, *N*>1, *M*>1	0.024	0.038	0.593	0.070	0.518	0.068
N:1	0.037	0.070	0.757	0.051	0.764	0.060
0:1	0.334	0.158	0.360	0.014	0.333	0.020
Total	1.000	1.000	0.444	0.034	0.477	0.040

Source: Computed from a digitized version of the Hand and Machine Labor study, see text and US Department of Labor (1899).
Notes: The unit of observation is a task as described by the staff of the Hand and Machine Labor study. The basic sample size in Panel A is 7,152 hand tasks from 610 production units. The basic sample size in Panel B is 12,473 machine labor tasks from 610 production units. 1:0 transitions are the hand labor tasks that disappeared in the transition to machine labor. 1:1 transitions are hand labor tasks that have a unique counterpart in machine production. In a 1:*M* transition, a single hand labor task subdivides into *M* machine tasks. In an *N*:*M* transition, *N* hand labor tasks transition into *M* machine labor tasks, with $N > 1$ and $M > 1$. In an *N*:1 transition *N* hand tasks combine to a single machine task. 0:1 indicates new tasks under machine labor. For equal weights, observations count equally in determining the average task shares within units. For time weights, observations are weighted by completion time in determining the average task shares within units.

were mechanized, whether by steam or water, similarly weighted. The sample used to compute Table 1 covers 610 of the original 626 manufacturing units.[6] For the most part, our discussion of Table 1 focuses on the equally weighted (rather than the time-weighted) statistics in Table 1.

Although some hand tasks were abandoned in the transition to machine labor, these comprised a small share of hand tasks and of the time spent in hand labor. The largest category of transitions by far was 1:1—that is, the agents were able to match a singleton task in hand production with a singleton task in machine production whose content was deemed to be the same, except that in machine production of the product, the task was far more likely to be mechanized. As can be seen from

[6] We excluded units from foreign countries, those that used horses, and which were otherwise missing data necessary for our analyses.

The American Divergence: Frontier vs. Coercion

- **The Free North (Habakkuk Thesis):** An open frontier made free labor scarce and expensive. High wages *induced innovation* specifically aimed at saving labor (e.g., the McCormick reaper, interchangeable parts).
- **The Coerced South (Slavery):** Violent coercion artificially suppressed labor costs to subsistence levels.
- **The Result:** With access to guaranteed cheap labor, Southern planters had little financial incentive to invest in labor-saving technology. The region lagged in mechanization while the North industrialized.

3. The East Asian Miracles: Induced Innovation

- Economies like Japan, South Korea, and Taiwan experienced different technological trajectories, supported by early, widespread land reform.
- **Induced Innovation:** Rather than importing large, labor-displacing Western factories wholesale, they adapted technology to their specific factor endowments (scarce land, abundant labor).
- They reverse-engineered Western designs to create smaller, appropriate machinery that could be utilized within extensive subcontracting networks.

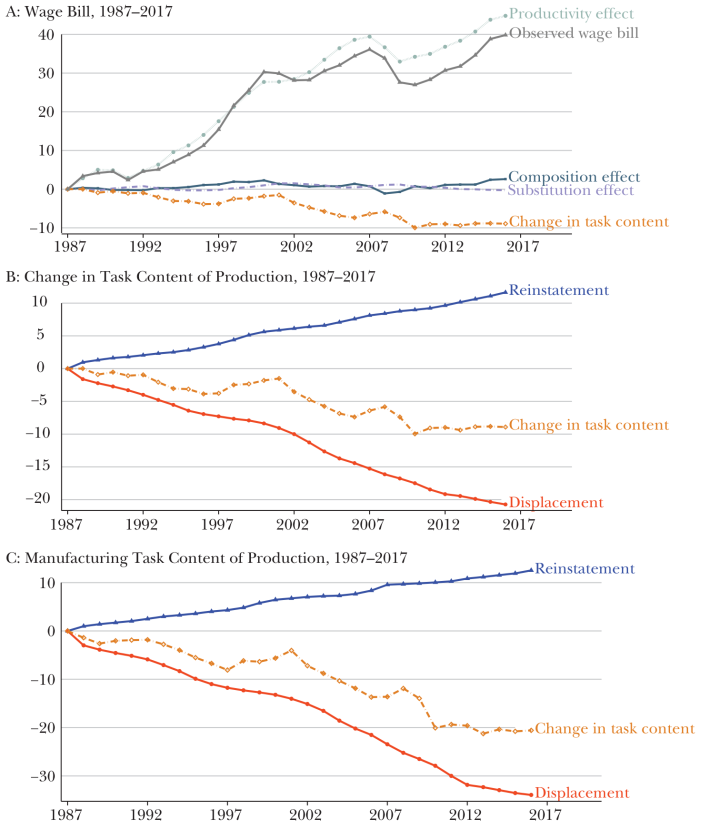
Expanding the Task Frontier

- By adapting technology in this way, these economies generated a substantial **Reinstatement Effect**, expanding labor-intensive tasks in light manufacturing.
- **Human Capital:** State investments in education ensured the workforce could adapt to these new, increasingly complex tasks.
- **Export Orientation:** Competing in global markets provided highly elastic demand.
- **The Result:** A strong Productivity Effect. Output scaled rapidly, allowing labor demand to keep pace with displacement and driving sustained wage growth.

The US Golden Age (1947–1987)

- Returning to the modern U.S. context, Acemoglu & Restrepo label the post-WWII era a "Golden Age."
- While automation displaced workers during this period, it was matched by a consistent **Reinstatement Effect** (the creation of new tasks).
- The net effect on the "task content of production" remained relatively stable.
- Consequently, wages and

Figure 5
Sources of Changes in Labor Demand, 1987–2017



Source: Authors' calculations.
Note: The top panel presents the decomposition of wage bill divided by population between 1987 and 2017. The middle and bottom panels present our estimates of the displacement and reinstatement effects for the entire economy and the manufacturing sector, respectively. See text for the details of the estimation of the changes in task content and displacement and reinstatement effects.

demand to decouple from productivity. Cumulatively, changes in the task content of production reduced labor demand by 10 percent during this period.

The middle and bottom panels of Figure 5 show that, relative to the earlier period, the change in task content is driven by a deceleration in the introduction

A Structural Shift (1987–2017)

- The data indicates a notable shift in recent decades.
- **Displacement** has continued, driven by software, algorithms, and robotics.
- However, the **Reinstatement Effect** has slowed significantly.
- The authors suggest an increase in "so-so" technologies: innovations that are just efficient enough to replace human labor but fail to generate the large productivity gains needed to boost broader labor demand

level, and then document the correlation between these measures and our estimates of the change in the task content of production.

We have three measures of industry-level automation technologies. The proxies are: 1) the adjusted penetration of robots measure from Acemoglu and Restrepo (2018b) for 19 industries, which are then mapped to our 61 industries; 2) the share of routine jobs in an industry in 1990, where we define routine jobs in an occupation as in Acemoglu and Autor (2011) and then project these across industries according to the share of the relevant occupation in the employment of the industry in 1990 (see also vom Lehn 2018); and 3) the share of firms (weighted by employment) across 148 detailed manufacturing industries using automation technologies, which include automatic guided vehicles, automatic storage and retrieval systems, sensors on machinery, computer-controlled machinery, programmable controllers, and industrial robots.[11]

Table 1 reports the estimates of the relationship between the change in task content of production between 1987 and 2017 and the proxies for automation technologies and new tasks; each row and column corresponds to a different regression model. The table shows that with all these proxies there is the expected negative relationship between higher levels of automation and our measure of changes in the task content of production in favor of labor (see also visual representations of these relationships in the online Appendix). These negative relationships remain very similar when we add various control variables, including, in column 1, a dummy for the manufacturing sector and, in column 2, imports from China (the growth of final goods imports from China as in Autor, Dorn, and Hanson 2013; Acemoglu, Autor, Dorn, Hanson, and Price 2016) and a measure of offshoring of intermediate goods (Feenstra and Hanson 1999; Wright 2014). Consistent with our conceptual framework, changes in task content are unrelated to imports of final goods from China, but are correlated with offshoring, which often involves the offshoring of labor-intensive tasks (Elsby, Hobijn, and Sahin 2013). Controlling for offshoring does not change the relationship we report in Table 1 because offshoring is affecting a different set of industries than our measures of automation (see the online Appendix).

We also looked at a series of proxies for the introduction of new tasks across industries, and how they are correlated with our measure of the change in task content for 1987–2017. Our four proxies for new tasks are: 1) the 1990 share of employment in occupations with a large fraction of new job titles, according to the 1991 *Dictionary of Occupational Titles* compiled by Lin (2011); 2) the 1990 share of employment in occupations with a large number of "emerging tasks" according to O*NET, which correspond to tasks that workers identify as becoming increasingly

[11] These data are from the Survey of Manufacturing Technologies, and are available in 1988 and 1993 for 148 four-digit SIC industries which are all part of the following three-digit manufacturing sectors: fabricated metal products; nonelectrical machinery, electric and electronic equipment; transportation equipment; and instruments and related products (Doms, Dunne, and Troske 1997). For this exercise, we computed measures for the change in task content of these four-digit manufacturing industries using detailed data from the Bureau of Economic Analysis input-output tables for 1987 to 2007.

Explaining the Current Trajectory

- If the direction of technology is shaped by incentives, why the shift toward labor displacement?
- **Institutional Bias:** The current U.S. tax code tends to subsidize capital investment while taxing human labor (via payroll taxes).
- **Industry Incentives:** The venture capital model often prioritizes software designed to substitute for labor and reduce immediate payroll costs, rather than the longer-term investments required to generate new human-centric industries.

Historical Summary

Regime	Institutional / Factor Context	Impact on Task Frontier	Net Outcome
Early Britain	State suppresses labor; enclosures increase supply.	Displacement heavily outpaces Reinstatement.	Wage stagnation (Engels' Pause).
U.S. Frontier	Land abundance creates high wages for free labor.	High wages induce automation; new factory tasks follow.	High productivity & wage growth.
Slavery	Coercion enforces artificial subsistence labor.	Stalls automation and new task creation.	Technological stagnation.
East Asia	Land reform + Export orientation.	Adaptation drives Reinstatement + Productivity.	Broad-based wage growth.

Concluding Thoughts

- The Industrial Revolution fundamentally reorganized the allocation of tasks between capital and labor.
- Economic history suggests that outcomes for workers are not strictly determined by the capabilities of new machines, but by the surrounding economic and political institutions.
- Sustained wage growth generally requires technology directed toward creating new tasks, supported by policies that invest in human capital and maintain labor's bargaining position.

Gemini is AI and can make mistakes.